

A COMBINED Z CROSS SECTION FROM CMS OPEN DATA

$Z \rightarrow \mu^+ \mu^-$ and $Z \rightarrow \tau_h \tau_h$, and what happens when you put them together

15 September 2026 • CMS Open Data 2016 G+H, $\sqrt{s} = 13 \text{ TeV}$, $L = 16.4 \text{ fb}^{-1}$

The result

- $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = 1931 \pm 35 \text{ pb}$ per lepton flavour
- the two channels agree: $\chi^2/\text{ndf} = 1.70/1$, $p = 0.19$
- lepton universality: $R = \sigma(\tau\tau)/\sigma(\mu\mu) = 1.21 \pm 0.17$
- **combining gains nothing** — the luminosity uncertainty is fully correlated and cannot be combined away

This deck

1. the result, and why it looks like that
2. strategy: two channels, one shared contract
3. $Z \rightarrow \mu\mu$ and $Z \rightarrow \tau_h \tau_h$ in two slides each
4. how the combination is actually done

+ backup: references, full breakdown, variations

The measurement at a glance

Data. CMS 2016 Open Data, Run2016G+H, NanoAODv9.
 $\sqrt{s} = 13 \text{ TeV}$, $L = 16\,393.381 \text{ pb}^{-1} \pm 1.2\%$ (normtag).

Two independent measurements, same luminosity, same simulation, same fitter (TRExFitter v1.8.0), same nuisance-parameter names:

- $Z \rightarrow \mu^+ \mu^-$: 10.4 M events, $\delta\sigma/\sigma = 1.8\%$
- $Z \rightarrow \tau_h \tau_h$: 47.6 k events, 80% of them jet $\rightarrow\tau_h$ fakes, $\delta\sigma/\sigma = 14\%$

Combination. Covariance (BLUE) of the two, with the shared systematics correlated through the Category contract of fitting/CONVENTIONS.md.

The answer, and the punchline: the two agree, and combining them gains **nothing** – because the luminosity uncertainty is fully correlated and irreducible.

Result

$$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = 1931 \pm 35 \text{ pb}$$

$$= 1931 \pm 0.6 \text{ (stat)} \pm 24 \text{ (syst)} \pm 12 \text{ (acc)} \pm 23 \text{ (lumi)} \text{ pb}$$

per lepton flavour, assuming lepton universality

$$\text{channel compatibility } \chi^2/\text{ndf} = 1.70/1, p = 0.19$$

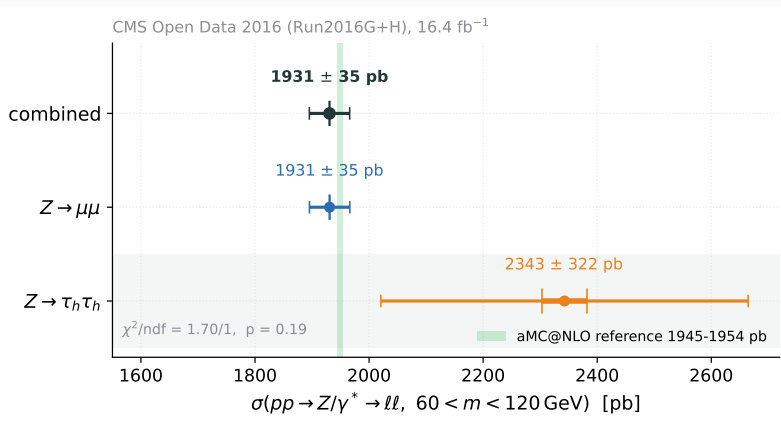
Lepton universality

$$R = \sigma(Z \rightarrow \tau\tau)/\sigma(Z \rightarrow \mu\mu) = 1.21 \pm 0.17$$

+1.3 σ from unity; the luminosity cancels exactly

CMS measures $1952 \pm 4 \pm 18 \pm 45 \text{ pb}$ for the same quantity (SMP-20-004). We agree within 0.4σ .

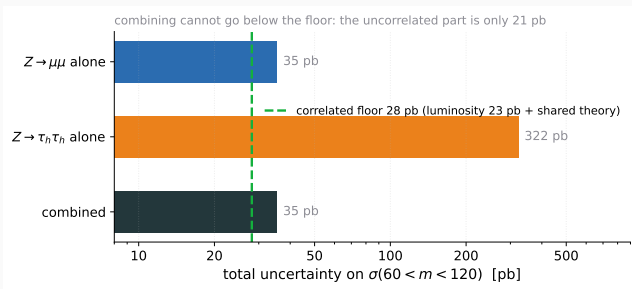
The combined cross section



Weights: $\mu\mu +1.0004$, $\tau\tau -0.0004$. A *negative* weight is normal BLUE behaviour once ρ exceeds the ratio of the uncertainties ($0.114 > 0.110$ — marginally, now that $\tau\tau$ has dropped its generator systematic): the imprecise measurement becomes a lever on the shared systematic, not an average.

The green band is the aMC@NLO reference, drawn as a *band* because the two channels normalise to 1945 and 1954 pb for the same quantity — a 0.47 % definitional difference we had to handle (backup).

Why combining gains nothing — and that is the result



$Z \rightarrow \mu\mu$ alone: 1930.7 ± 35.36 pb.

Combined: 1930.6 ± 35.36 pb.

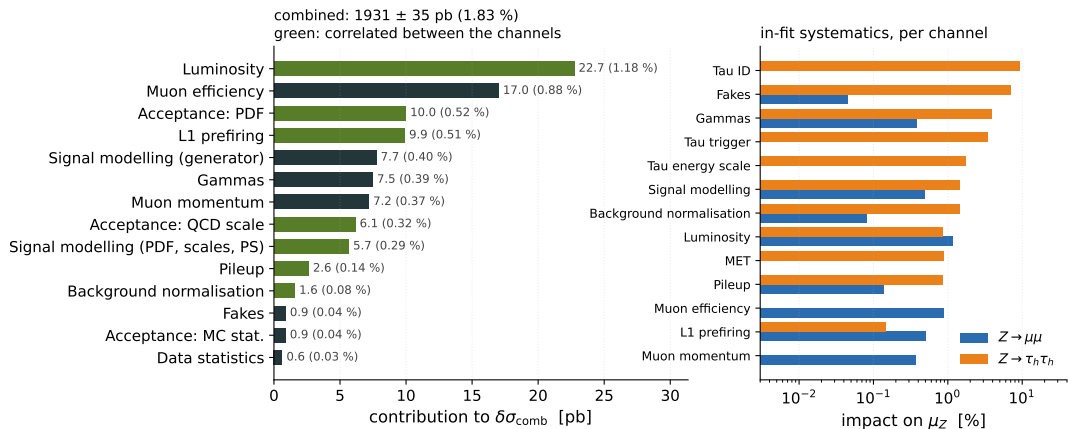
Improvement: 0.00 %.

Why. 23 pb of the 35 pb is the luminosity: identical normtag value, identical runs, identical $\pm 1.2\%$ in both channels. A fully correlated uncertainty cannot be combined away. Add the shared theory terms and the floor is 28 pb.

A third channel of the same precision would not help either. The route to a better Z cross section from this dataset is a **better luminosity calibration**, not more channels.

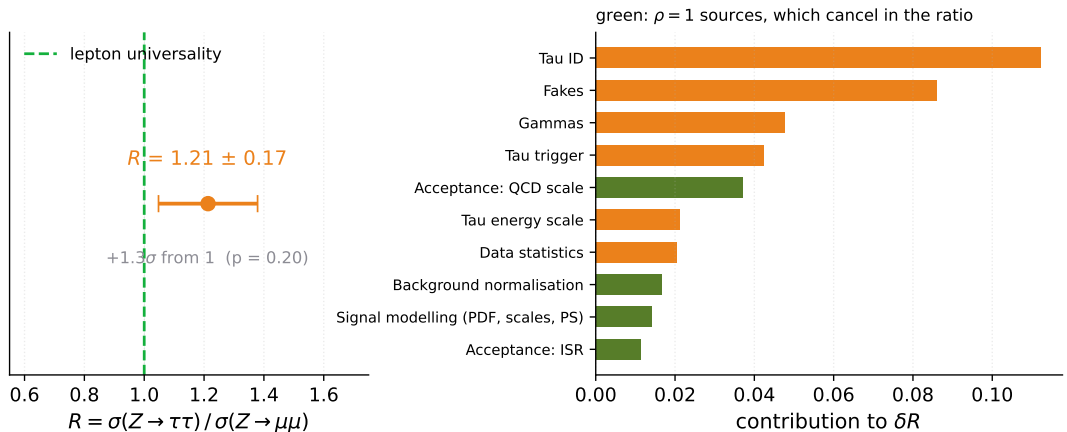
Data statistics contribute 0.6 pb – 0.03 %. This measurement was never going to be limited by the number of collisions.

Where the uncertainty comes from



Left: contribution of each source to the combined uncertainty ($\delta_s = \sqrt{w^T C_s w}$, adding in quadrature to the total); green = correlated between the channels. Right: the same systematics as each channel's fit reports them. The $\tau\tau$ channel's τ_h ID (11.1 % on its own μ_Z) survives into the combination as 0.08 pb.

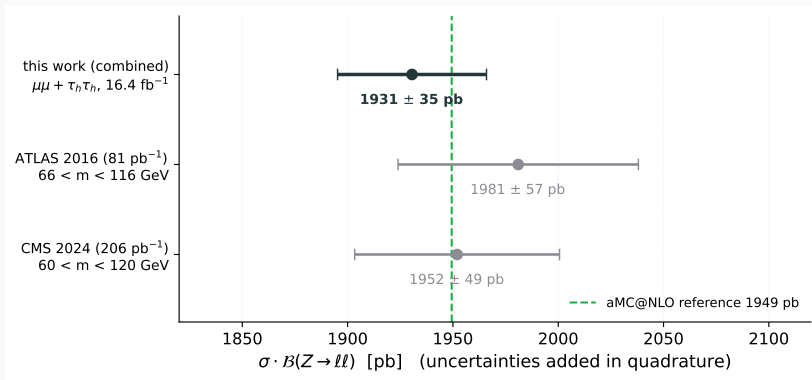
Lepton universality — what the $\tau\tau$ channel actually buys



The combination *assumes* universality, so it cannot test it. The ratio can — and in the ratio every $\rho = 1$ source cancels exactly: luminosity, pileup, L1 prefire, and the shared PDF/scale/shower terms.

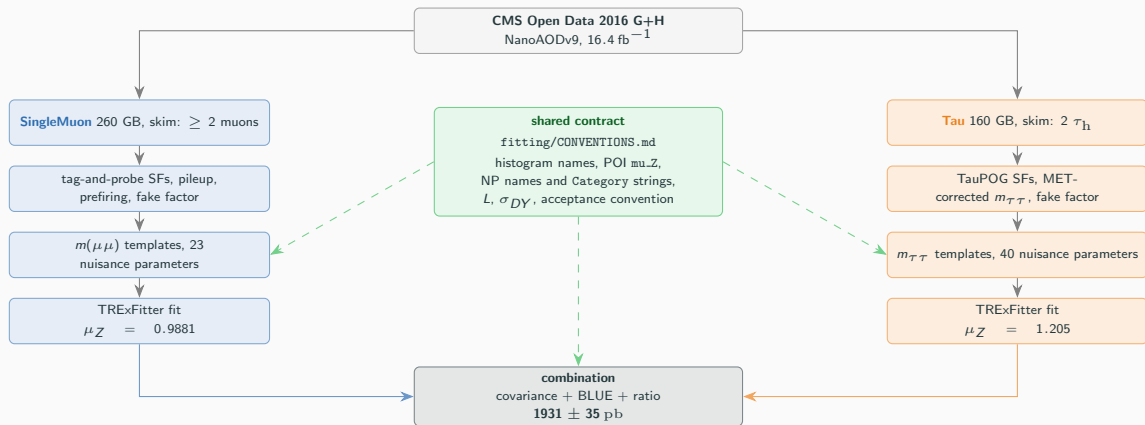
What is left is the $\tau\tau$ channel's own calibration: τ_h ID (11.1 %) and the data-driven fake factor (8.6 %). Improving those improves the *universality test*; it does essentially nothing for the cross section.

Comparison with published measurements



Agreement with the CMS measurement of exactly the same quantity is within 0.4σ ; the ATLAS point uses a narrower window (66–116 GeV) and is shown for scale only. Both published numbers carry a *larger* relative luminosity uncertainty than we do (2.3% and 2.1% against 1.2%) — they are early-Run-2 measurements, while the 2016 G+H norntag luminosity is well measured. Note this is not a fully independent check: the same NNLO reference ($\sigma_{DY} = 6077.22 \text{ pb}$) normalises the simulation behind both of our C factors.

How the three pieces fit together



The **green** contract is what makes the combination possible at all: because both channels name 'Lumi', 'Pileup', 'L1Prefiring', 'PDF' ... identically and group them into the same Category, the correlations can be assigned category by category instead of guessed.

Data and simulation

Collision data (Run2016G+H only — 16 of the 36 fb^{-1} CMS recorded in 2016)

dataset	recid	events
SingleMuon G / H	30530 / 30563	324 M
Tau G / H	30532 / 30565	156 M

Certified with the 2016 golden JSON restricted to runs 278820–284044.

Luminosity. $L = 16\,393.381\text{ pb}^{-1}$, the *normtag* value from CMS Open Data record 1059. An earlier *brilcalc* call without `--normtag` gave $16\,290.713\text{ pb}^{-1}$, 0.63 % low; both channels now use the *normtag* value, which is why the luminosity is *exactly* correlated between them.

Simulation (UL16 postVFP, streamed from EOS or read from dCache)

sample	recid	used for
DY M-50 aMC@NLO	35669	signal, both
DY M-50 madgraph	35671	generator syst.
DY M-10to50	35631	$\tau\tau$ background
$t\bar{t}$ 2L2Nu / semilep.	67801 / 67993	background
tW top / antitop	64895 / 64839	background
WW, WZ, ZZ	72696/72754/75593	background
W +jets aMC@NLO	69745	$\tau\tau$ background

Normalisation. $\sigma(Z/\gamma^* \rightarrow \ell\ell, m > 50\text{ GeV}) = 6077.22\text{ pb}$ (NNLO), divided by 3 per flavour. Both channels' μ_Z is a multiplier on *that* number, which is what makes them comparable — and why the residual 0.47 % mismatch matters (backup).

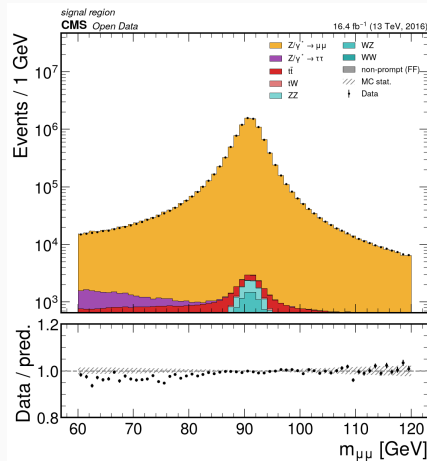
$Z \rightarrow \mu^+ \mu^-$: a 1.8 % measurement with 10 million events

Selection. HLT_IsoMu24 || HLT_IsoTkMu24; exactly two tight, isolated, opposite-sign muons, $p_T > 26/20$ GeV, $|\eta| < 2.4$, $60 < m(\mu\mu) < 120$ GeV; FSR photons recovered from the FsrPhoton collection.

Corrections, all measured in this data.

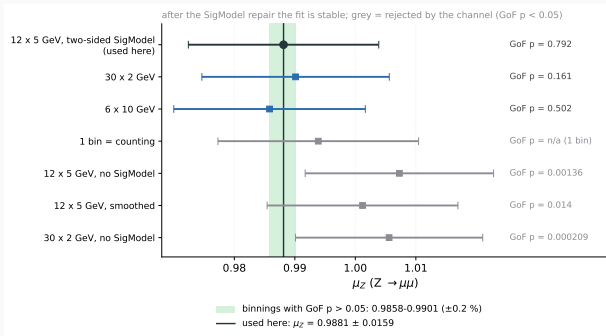
- ID / isolation / trigger scale factors from tag-and-probe fits on the Z peak (20 M pairs), with trigger-object matching
- pileup weights from the Open Data luminosity table, L1-prefiring weights
- Z-peak muon momentum calibration per $|\eta|$ bin
- non-prompt muons from a same-sign fake factor (3.8 k events, 0.04 % of the yield)

Backgrounds are tiny: 69 k of 10.4 M events (0.66 %), dominated by $t\bar{t}$ (32 k) and $Z \rightarrow \tau\tau$ (11 k).



The distribution that is fitted, shown in 1 GeV input bins; the fit runs in 12×5 GeV. Note the ratio panel — data sit 3–4 % below the prediction at 62–78 GeV. That deficit is the reason the next slide exists.

$Z \rightarrow \mu\mu$: which number to take — the channel's review, and its repair



The channel's own review (`z-mumu/REVIEW.md`, F3/F4) found the 30-bin shape fit **not robust**: μ_Z moved over 0.990–1.006, because the only template that could absorb a 3–4 % data deficit at 62–78 GeV was SigModel — and that template mixed the generator comparison with a phase-space artefact (the powheg sample is generated with $m_{\text{LHE}} < 120$ GeV, depleting its two highest bins by 10 % and 33 %).

The channel fixed both. Fit in 12×5 GeV bins so the shape parameters are not over-constrained, no smoothing, MINOS everywhere; SigModel rebuilt with both generators inside a common $50 < m_{\text{LHE}} < 120$ GeV window and mirrored.

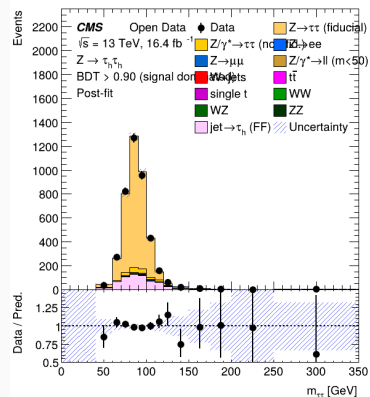
We therefore take the shape fit: $\mu_Z = 0.9881 \pm 0.0159$, GoF $p = 0.79$, stable to $\pm 0.2\%$ across the binnings that describe the data. The $\pm 0.7\%$ lineshape term the review had asked for as a stop-gap is **retired** — the two-sided template carries it inside the fit. Counting instead would move the combination by +11.2 pb: reported as a variation, not used.

$Z \rightarrow \tau_h \tau_h$: a 14 % measurement where 87 % of the data is background

Selection. Double- τ_h trigger (different path in G and H), both legs HLT-matched; two opposite-sign τ_h with $p_T > 40$ GeV, $|\eta| < 2.1$, decay modes 0/1/10/11, DeepTau2017v2p1 Medium vs jets; e/μ veto keeps it orthogonal to the other channels. $\Rightarrow$ 47 586 events, of which $\sim 80\%$ are jets faking a τ_h .

No QCD simulation is used. The fake background comes from data: a fake factor $FF(\text{era}, DM, N_{\text{jets}}, p_T)$ measured in a same-sign region and applied to opposite-sign events whose leading τ_h fails Medium, times an OS/SS correction of 1.04–1.15 measured per (era, N_{jets} , BDT category) in the anti-isolated sideband — new in v2.1, together with a k -fold BDT (held-out AUC 0.970) that splits the signal region into three categories by signal purity.

Mass. The visible di- τ mass is not a Z mass. A per-event likelihood over the τ visible-energy fractions using the MET covariance gives $m_{\tau\tau}$ with scale 0.99 and resolution 11 % (against 0.80 / 13 % for m_{vis}).



Post-fit $m_{\tau\tau}$ in the most signal-like BDT category (score > 0.90 , $S/B \approx 5.3$); the fit runs over all three, 14 bins each, 0–350 GeV. Pink = the data-driven jet $\rightarrow \tau_h$ fakes, orange = $Z \rightarrow \tau\tau$. Counting over the whole signal region instead gives 2447 pb at $B/S \approx 8$: it is the low-score categories that fix the fake normalisation.

$Z \rightarrow \tau_h \tau_h$: result and what limits it

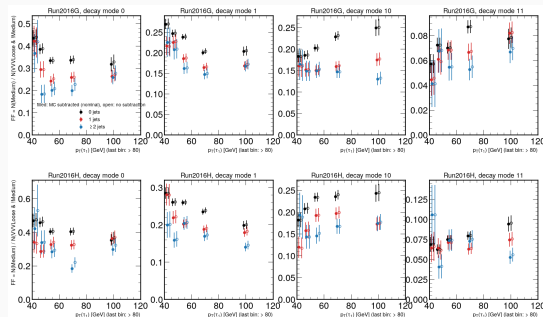
$$\mu_Z = 1.205^{+0.145}_{-0.125}$$

$$\sigma(60\text{--}120) = 2343 \pm 39^{+282}_{-243} \pm 86 \text{ pb}$$

goodness of fit $p = 0.22$

source	impact on μ_Z
τ_h ID	11.1 %
fakes (fake factor)	8.6 %
MC statistics	4.7 %
τ_h trigger	4.2 %
τ_h energy scale	2.1 %
signal modelling (PDF, scales, PS)	1.8 %
luminosity	1.0 %
data statistics	2.0 %

Acceptance $A = 0.23\%$ with a 3.7% uncertainty — the double 40 GeV cut on the *visible* τ momentum sits on the Z p_T tail, where the QCD scale dependence is 3.4% .



The fake factors, binned in era, decay mode, N_{jets} and p_T . Without the N_{jets} binning the same-sign closure is off by 10–25 % per jet bin; without the era binning by $\pm 5\%$ (the HLT τ isolation changed between G and H).

What is combined, and how

The observable.

$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV})$ per lepton flavour, assuming lepton universality:

$$\sigma_c = \hat{\mu}_c \times \sigma_c^{\text{pred}}(60-120)$$

Not μ_Z itself: the two channels normalise to references 9 pb apart for the same quantity (backup), so averaging the μ 's would average two different things.

The covariance, source by source,

$$C = \sum_s C_s, \quad (C_s)_{ij} = \rho_s s_i s_j$$

then standard BLUE,

$$w = \frac{C^{-1}u}{u^T C^{-1}u}, \quad \hat{\sigma} = w^T x, \quad \text{Var} = \frac{1}{u^T C^{-1}u}$$

with χ^2 (ndf=1) for compatibility, and the combined uncertainty decomposed as $\delta_s = \sqrt{w^T C_s w}$.

Two details that are easy to get wrong.

In-fit systematics scale with the prediction, because μ_Z multiplies a fixed number: $\delta\sigma = s_{\text{group}} \times \sigma^{\text{pred}}$. *The acceptance is multiplicative*, $\delta\sigma = a \times \sigma^{\text{meas}}$, since $\sigma = \sigma_{\text{fid}}/A$ — and A is outside both likelihoods (0.61 % for $\mu\mu$, 3.7 % for $\tau\tau$), so the combination has to add it.

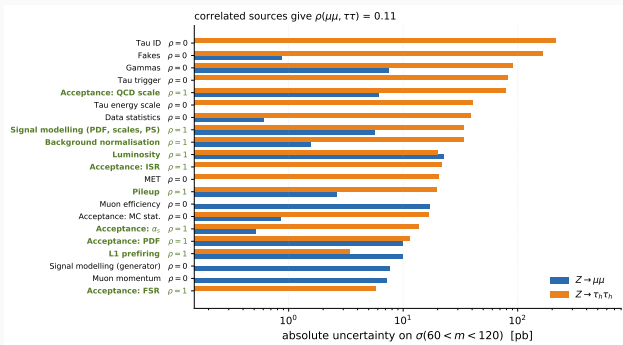
Because the acceptance is multiplicative, the BLUE is *iterated*: the terms are re-evaluated at the combined value until stable (4 iterations, < 0.1 pb).

Cross-checked against an explicit profile likelihood with one nuisance per shared source,

$$-2 \ln L = \sum_c \frac{(x_c - \sigma - \sum_s d_{cs} \theta_s)^2}{u_c^2} + \sum_s \theta_s^2$$

which additionally carries the $\tau\tau$ channel's asymmetric error. Same answer: 1930^{+35}_{-36} against 1931 ± 35 pb.

The correlation model — the only judgement in the whole thing



$\rho = 1$ — derived once, used by both

- Luminosity: same normtag value, same runs, same $\pm 1.2\%$
- Pileup, L1 prefiring: same profile, same maps, same code
- Background normalisation: literally the same XS_* parameters on the same MC samples
- PDF, α_s , QCD scale, PS: same Hessian members and the same 7-point envelope of the *same* aMC@NLO sample

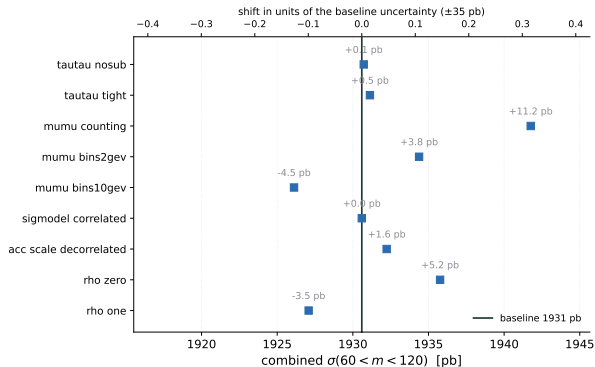
$\rho = 0$ — different objects, methods or events

- muon vs τ_h calibrations: no common input
- Gammas: same sample, but $\mu\mu$ and $\tau_h\tau_h$ select disjoint events
- fakes, MET
- data statistics: *disjoint* primary datasets, orthogonal triggers

The split of Signal modelling. Both channels used to fit a generator systematic called SigModel — but $\mu\mu$ compares two NLO generators (0.4% on μ_Z) and $\tau\tau$ compared LO with NLO. In v2.1 $\tau\tau$ **dropped** its version: it is reported, not fitted, because “nominal minus a worse model” is a placeholder that double-counts QCDscale/PS_ISR. We still split the category and decorrelate the generator part; on the $\tau\tau$ side it is now zero, so the “SigModel correlated” variation is exactly null.

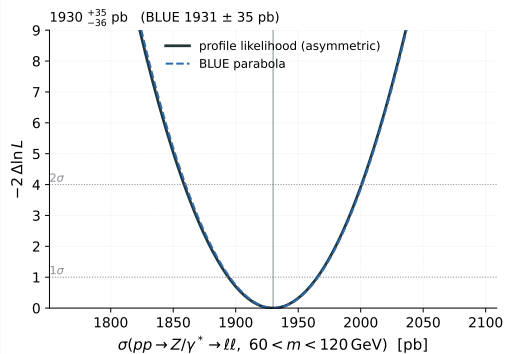
$\Rightarrow \rho(\mu\mu, \tau\tau) = 0.114$ — it was 0.15 before $\tau\tau$ dropped that term.

Cross-checks: the model is not what limits us



Every variation moves the answer by less than 0.35σ . The two rows that *bracket the entire correlation model* — all systematics uncorrelated, all fully correlated — span 8.7 pb, a quarter of the total uncertainty. The largest single variation is not a correlation choice at all: it is which $\mu\mu$ extraction to take (+11.2 pb for counting instead of the fit). The two $\tau\tau$ rows move it by 0.1 and 0.5 pb.

Right: the profile likelihood and the BLUE parabola lie on top of each other — the honest way of saying that the $\tau\tau$ channel's asymmetry does not matter here.



What this is not, and what is left to do

Not a TRExFitter MultiFit

The channel workspaces are not committed to the repository and were not rebuilt, so the shared nuisance parameters are correlated through their Category totals rather than *profiled jointly*. The fitting is still TRExFitter's — it produced both inputs.

The MultiFit config is generated and syntax-checked (`combination/fit/comb.config`) and one command away once the workspaces exist:

```
git submodule update --init --recursive
bash fitting/build_trexfitter.sh
cd z-mumu && python run_v2.py --from 4
cd ../z-tautau && python run_all.py --from 3
cd ../combination && python fit/run_multifit.py
```

What the joint fit would add

- shared NPs profiled together: $\mu\mu$ data constraining Lumi, Pileup, PDF ... for $\tau\tau$ as well
- τ_h ID constrained *in situ* if $e\tau_h/\mu\tau_h$ were added
- per-bin γ parameters instead of one Gammas number
- the $\tau\tau$ asymmetry exactly

Expected to be small ($\mu\mu$ carries essentially all the weight and is already profiled), but it is not measured, so the claim is not made.

Inherited caveats we did not fix

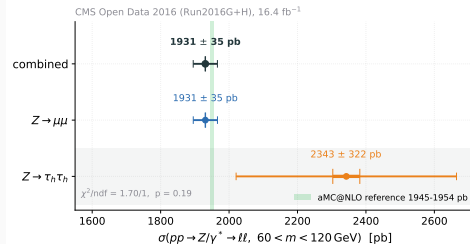
- $\mu\mu$: the 60–80 GeV lineshape lies between aMC@NLO and powheg, and the fit resolves it with SigModel $+0.66\sigma$; prefiring, muon and pileup calibrations all in-house
- $\tau\tau$: DeepTau *Tight* is the channel's recommended next working point ($+0.5$ pb, $R = 1.08 \pm 0.13$) but is not yet its nominal
- no $Z \rightarrow ee$ — out of scope for this pass

Conclusions

$$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = 1931 \pm 35 \text{ pb}$$

$$R = \sigma(Z \rightarrow \tau\tau)/\sigma(Z \rightarrow \mu\mu) = 1.21 \pm 0.17$$

1. Two very different analyses of the same 16.4 fb^{-1} — 10 M nearly background-free events against 47 k events that are 87 % background — **agree**.
2. The combination is **luminosity-limited**: 23 pb of the 35 pb, fully correlated, irreducible. Combining gains nothing at all (0.00 %).
3. Data statistics are 0.6 pb. **More collisions would change nothing**; better calibration and more simulated events would.
4. What the $\tau\tau$ channel really contributes is the **universality test**, where the luminosity cancels exactly.
5. The shared conventions made this possible at all — and the places where they were ambiguous (SigModel, the reference cross section) are exactly where the work went.



All numbers, code and documentation: `combination/result.md`,
`combination/docs/`. Reproduces from git in seconds with `python`
`run_combination.py`.

Backup

Backup: the two references differ by 0.47 %

Both channels define μ_Z as a multiplier on an aMC@NLO prediction of $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV})$ — but they build it differently:

channel	construction	value
$\mu\mu$	$\sigma_{\text{fid}}^{\text{pred}} / A(60\text{--}120)$ 799.566/0.409209	= 1953.9 pb
$\tau\tau$	LHE- $\tau\tau$ subset of the same sample, $60 < m_{\text{LHE}} < 120$	1944.9 pb

With 72 M generated events the statistical precision on either is 0.02 %, so the 0.47 % is **definitional**, not a fluctuation.

Why it matters.

- We combine *cross sections*: each $\hat{\mu}$ is multiplied by its own reference first, so the mismatch shows up as a band on the theory line rather than as a bias.
- A TRExFitter MultiFit with one shared μ_Z would *not* handle it — it would fit one parameter against two references.
- Size today: the $\tau\tau$ weight is 0.04 %, so ~ 0.0002 % on the combination. It matters as soon as a second channel of comparable weight appears.

Recorded as a decision in `fitting/CONVENTIONS.md` §6.

Backup: full uncertainty breakdown of the combined value

source	pb	of σ	corr.
Luminosity	22.72	1.18 %	yes
Muon efficiency	17.00	0.88 %	no
Acceptance: PDF	9.97	0.52 %	yes
L1 prefire	9.89	0.51 %	yes
Sig. modelling (generator)	7.74	0.40 %	no
Gammas (MC stat.)	7.46	0.39 %	no
Muon momentum	7.17	0.37 %	no
Acceptance: QCD scale	6.15	0.32 %	yes
Sig. modelling (PDF, scales, PS)	5.65	0.29 %	yes
Pileup	2.61	0.14 %	yes
Background normalisation	1.57	0.08 %	yes
Fakes	0.87	0.04 %	no
Acceptance: MC stat.	0.86	0.04 %	no
Data statistics	0.61	0.03 %	no
Acceptance: α_s	0.52	0.03 %	yes
Tau ID	0.08	0.00 %	no
Tau trigger	0.03	0.00 %	no
Tau energy scale	0.02	0.00 %	no
total	35.4	1.83 %	

Contributions add in quadrature: $\delta_s = \sqrt{w^T C_s w}$ with C_s the covariance of that source alone.

Reading the table:

- the top entry and three of the top seven are **correlated**, so they set a floor combining cannot touch;
- the second entry is the $\mu\mu$ muon efficiency, 17 pb — in-house tag-and-probe scale factors plus the 0.8%/event reconstruction term;
- everything the $\tau\tau$ channel contributes sums to 0.2 pb;
- the data statistical uncertainty is 0.6 pb, four orders of magnitude below the cross section.

Generated table: `combination/result.md`, machine-readable version
`combination/output/combination.result.json`.

Backup: variations in numbers

variation	σ [pb]	shift	what it changes
baseline	1930.6	—	$\mu\mu$ shape fit, $\tau\tau$ mcsb, model of §4
$\tau\tau$ nosub	1930.7	+0.1	fake factor without the genuine- τ subtraction
$\tau\tau$ Tight WP	1931.1	+0.5	DeepTau Tight on both legs (the channel's next WP)
$\mu\mu$ counting	1941.8	+11.2	1-bin extraction instead of the 12×5 GeV fit
$\mu\mu$ 30×2 GeV	1934.4	+3.8	alternative binning (GoF $p = 0.16$)
$\mu\mu$ 6×10 GeV	1926.1	-4.5	alternative binning (GoF $p = 0.50$)
SigModel correlated	1930.6	+0.0	generator comparison shared — null, see §4
acc. scale decorrelated	1932.2	+1.6	acceptance QCD-scale term independent
$\rho = 0$ everywhere	1935.8	+5.2	all systematics uncorrelated (naive)
$\rho = 1$ everywhere	1927.1	-3.5	all systematics fully correlated

The $\rho = 0$ and $\rho = 1$ rows bracket every possible correlation assignment: they span 8.7 pb, 0.25σ . The single largest entry is not a correlation choice at all but the $\mu\mu$ signal extraction: counting instead of fitting moves the answer by 11.2 pb, and the two binnings the channel accepts span 8.3 pb. The two $\tau\tau$ rows move it by 0.1 and 0.5 pb — that channel cannot move this number, whatever it does.

Backup: the channels' own numbers

	$Z \rightarrow \mu^+ \mu^-$	$Z \rightarrow \tau_h \tau_h$
selected events	10 378 567	47 586
background	68 787 (0.66 %)	41 513 (87 %, prefit)
A (fiducial / 60–120)	0.4092	0.002314
C (reco / fiducial)	0.7914	0.0654
fit variable	$m(\mu\mu)$, 12×5 GeV	$m_{\tau\tau}$, 3×14 bins, 0–350 GeV
nuisance parameters	23 (+12 γ)	40 (+42 γ)
μ_Z	0.9881 ± 0.0159	$1.205^{+0.145}_{-0.125}$
σ_{fid}	790.1 pb (pred. 799.6)	5.42 pb (pred. 4.50)
$\sigma(60\text{--}120)$	1931 ± 35 pb	2343 ± 322 pb
goodness of fit	$p = 0.79$	$p = 0.22$

The asymmetry is the point: one channel is a precision measurement whose entire difficulty is calibration, the other is a needle-in-a-haystack whose entire difficulty is the background. They share a luminosity, a signal sample and a fitter, and nothing else.

Backup: where everything lives

combination/result.md	the result, generated from the JSON — never edited by hand
combination/docs/00--04	overview, inputs, correlation model, method, results
combination/comb/	inputs, model, blue, likelihood, ratio, plots, report
combination/run_combination.py	orchestrator; --check runs 6 closure tests
combination/fit/comb.config	the TRExFitter MultiFit, generated and validated, not run
combination/output/	combination_result.json, plots/*.pdf

z-mumu/, z-tautau/	the channel analyses, each with its own README, docs/, handoff.md
z-mumu/REVIEW.md	the review, and the fixes that made the shape fit usable
fitting/CONVENTIONS.md	the shared contract; §6 records what the combination decided
combination/presentation/	this deck (bash build.sh)

Closure tests behind --check: each published channel number is re-asserted; a single-channel BLUE reproduces that channel exactly; two identical measurements with $\rho = 0$ give error/ $\sqrt{2}$ and equal weights; with every systematic correlated only the statistical part averages; and the profile likelihood reproduces BLUE to 10^{-3} pb.